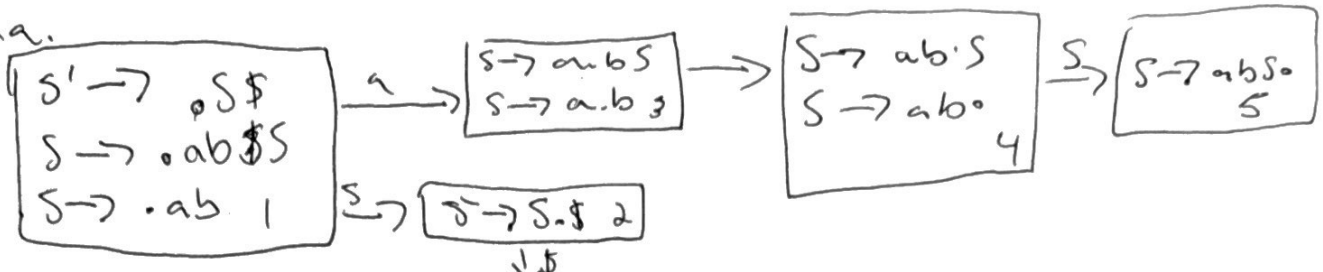
HW1

Stack	Stream	Action
ε_1	((id), id)$	S3
$\varepsilon_1(_3$	(id), id)$	S3
$\varepsilon_1(_3(_3$	id), id)$	S2
$\varepsilon_1(_3(_3 id_2$	), id)$	Reduce(S→id)
$\varepsilon_1(_3(_3 S_8$	), id)$	g7
$\varepsilon_1(_3(_3 S_{11}$	), id)$	reduce L→S
$\varepsilon_1(_3(_3 L$	), id)$	g5
$\varepsilon_1(_3(_3 L_5$	), id)$	S6
$\varepsilon_1(_3(_3 L_5)_6$	, id)$	reduce S→(L)
$\varepsilon_1(_3 S$	, id)$	g7
$\varepsilon_1(_3 S_7$	, id)$	reduce L→S
$\varepsilon_1(_3 L$	, id)$	g5
$\varepsilon_1(_3 L_5$	, id)$	S8
$\varepsilon_1(_3 L_5,_8$	id)$	S2
$\varepsilon_1(_3 L_5,_8 id_2$	)$	reduce S→id
$\varepsilon_1(_3 L_5,_8 S$	)$	g9
$\varepsilon_1(_3 L_5,_8 S_9$	)$	reduce L→L,S
$\varepsilon_1(_3 L$	)$	g5
$\varepsilon_1(_3 L_5$	)$	S6
$\varepsilon_1(_3 L_5)_6$	$	reduce S→(L)
$\varepsilon_1 S$	$	g4
$\varepsilon_1 S_4$	$	Accept

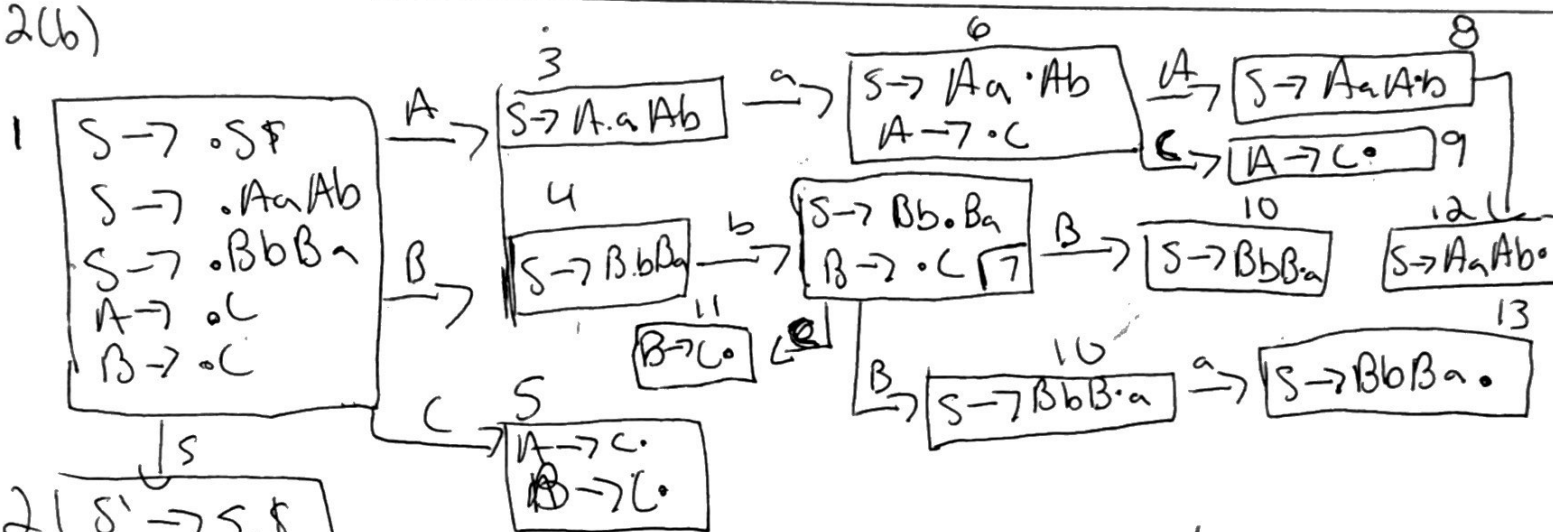
2a.



State	a	b	$	S
1	S_2			g_2
2			Accept	
3		S_3		
4				
5	S_2		$S \to ab$ $S \to abS$	g_5

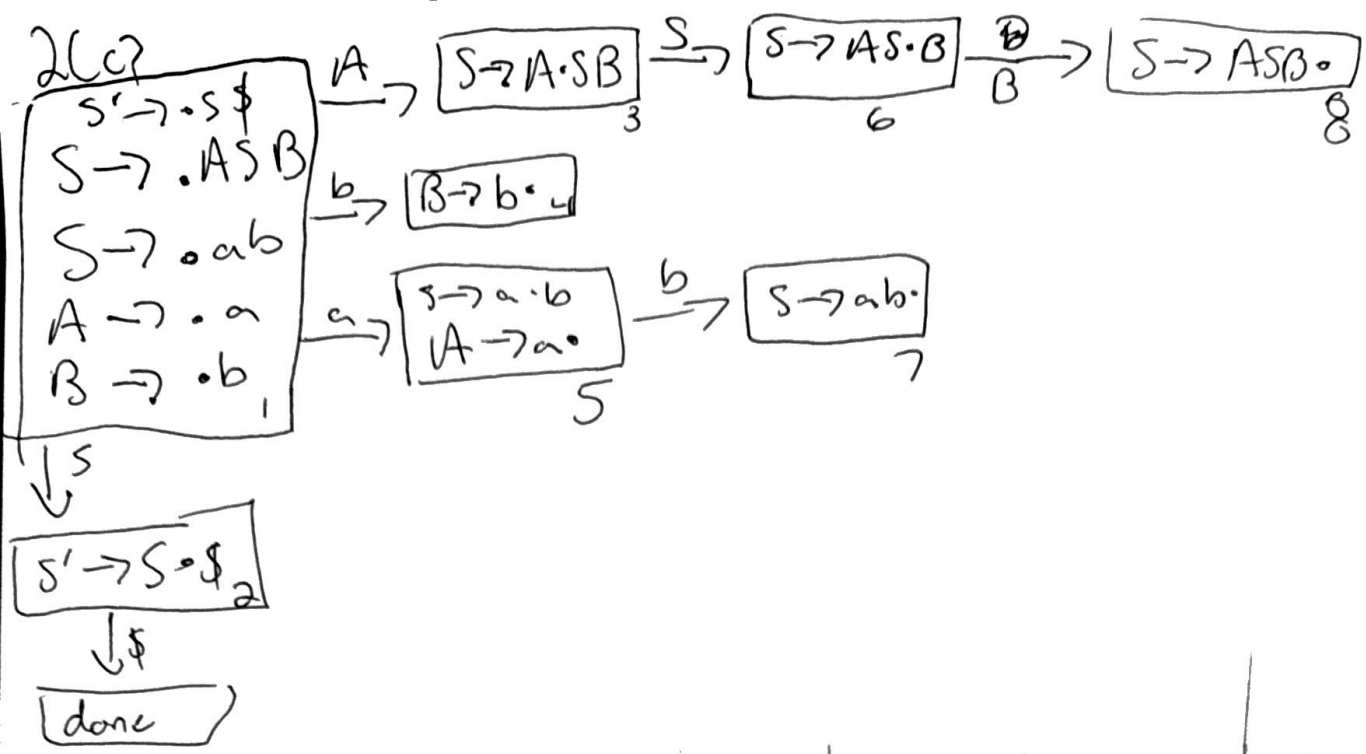
Yes, the grammar is SLR
there are no conflicts.

2(b)



State	a	b	c	$	S	A	B
1			S_5		g_2	g_3	g_4
2				Accept			
3	S_6						
4		S_7					
5	$A \to C. / B \to C.$			$A \to C. / B \to C.$			
6			S_9			g_6	
7			S_{11}				g_{10}
8		S_{12}					
9	$A \to C$	$A \to C$	$A \to C$	$A \to C$			
10	S_{13}						
11	$B \to C$	$B \to C$	$B \to C$	$B \to C$			
12				$S \to AaAb$ $S \to BbBa$			
13							

this grammar is not SLR because of state 5.



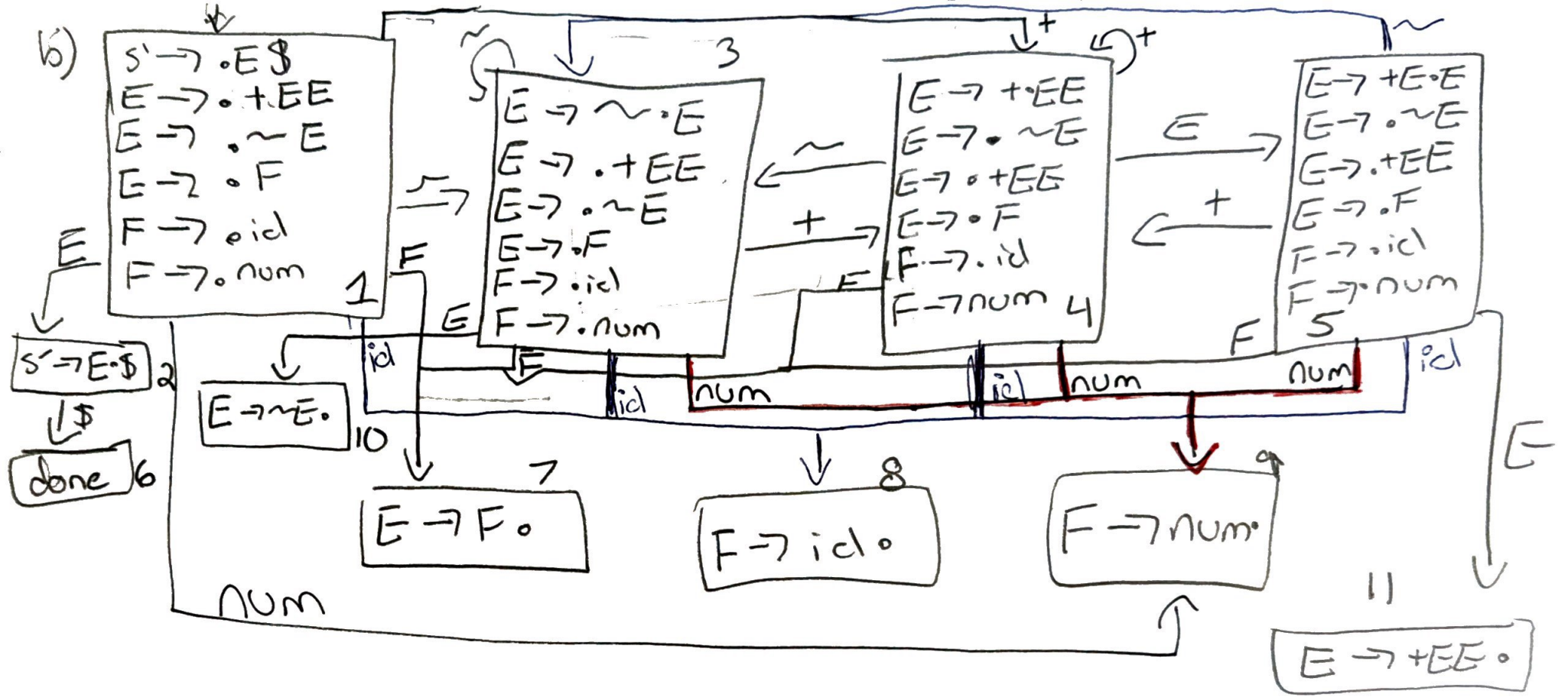
State	a	b	$	S	A	B
1	S5					
2	S'→S	S'→S	done	S'→S	S'→S	S'→S
3				g6		
4	B→b	B→b	B→b	B→b	B→b	B→b
5		S7				
6						g8
7	S→ab	S→.b	S→.b	S→ab	S→ab	S→ab
8	S→ASB	S→ASB	S→ASB	8→ASB	S→ASB	8→ASB
9			Accept			

The grammar is SLR since there are no conflicts.

3a. ++~512 eg

form	Rule Applied
1. S'	
2. E	S' → E
3. +EE	E → +EE
4. +EF	E → F
5. +E id	F → id
6. ++EE id	E → +EE
7. ++EF id	E → F
8. ++E num id	E → NUM
9. ++~E num id	E → ~E
10. ++~F num id	E → F
11. ++~num num id	F → num

++~ 5 12 eg


b)
S' → .E$
E → .+EE
E → .~E
E → .F
F → .id
F → .num
1
S' → E.$
2
done
6
E → ~E.
10
E → F.
7
E → ~.E
E → .+EE
E → .~E
E → .F
F → .id
F → .num
3
E → +.EE
E → .~E
E → .+EE
E → .F
F → .id
F → num
4
E → +E.E
E → .~E
E → .+EE
E → .F
F → .id
F → .num
5
F → id.
8
F → num.
9
E → +EE.
11
num
id
num
id
num
id
E
F
E
F
E
F
~
+
~
+
+
~
+
E
+
E

c).

first set

S'	E	F
+	+	id
~	~	num
id	id	
num	num	

follow set

S'	E	F
$	+	+
	~	~
	id	id
	num	num
	$	$

rule 1 for S'

rule 3 for E to get $

E → +EE first E is followed by second E

rule 2 | E → F so rule 3 for F

d).

State	+	~	id	num	$	E	F
1	s4	s3	s8	s9		g2	g7
2					done		
3	s4	s3	s8	s9		g10	g7
4	s4	s3	s8	s9		g5	g7
5	s4	s3	s8	s9		g11	g7
6	r1	r1	r1	r1	r1		
7	r4	r4	r4	r4	r4		
8	r5	r5	r5	r5	r5		
9	r6	r6	r6	r6	r6		
10	r3	r3	r3	r3	r3		
11	r2	r2	r2	r2	r2		

e) Flex

```
%%
[ \t\r\n]+ { }
"+"        { return PLUS; }
"~"        { return NEG; }
[a-z]+     { int n = 0;
             for(int i = 0; i < yyleng; i++){
               n = n * 26 + (yytext[i] - 'a');
             }
             yylval = n;
             return ID; }
[0-9]+     { yylval = atoi(yytext); return NUM; }
%%
```

Bison

```
%%
S : E      { printf("%d\n", $1); }
;
E : PLUS E E   { $$ = $2 + $3; }
  | NEG E      { $$ = -$2; }
  | F          { $$ = $1; }
;
F : ID     { $$ = $1; }
  | NUM    { $$ = $1; }
;
%%
```

f)

Stack	Steam	Action
ε_1	++ 512 cy $	S4
$\varepsilon_1 +_4$	+ 5 12 cy $	S4
$\varepsilon_1 +_4 +_4$	5 12 cy $	S9
$\varepsilon_1 +_4 +_4 5_9$	12 cy $	r6) id.
$\varepsilon_1 +_4 +_4 num_9$	12 cy $	r6
$\varepsilon_1 +_4 +_4 F_7$	12 cy $	r4
$\varepsilon_1 +_4 +_4 E_5$	12 cy $	S9
$\varepsilon_1 +_4 +_4 E_5 12_9$	cy $	r6
$\varepsilon_1 +_4 +_4 E_5 num_8$	cy $	r6
$\varepsilon_1 +_4 +_4 E_5 F_7$	cy $	r4
$\varepsilon_1 +_4 +_4 E_5 E_{11}$	cy $	r2
$\varepsilon_1 +_4 E_5$	Cy$	S8
$\varepsilon_1 +_4 E_5 cy_8$	$	r5
$\varepsilon_1 +_4 E_5 id_8$	$	r5
$\varepsilon_1 +_4 E_5 F_7$	$	r4
$\varepsilon_1 +_4 E_5 E_{11}$	$	r2

F)

Stack	Stream	Action
$\varepsilon_1 E_2$	$	g_2
$\varepsilon_1 E_2 \$$		done

G).

the parse in F) is the rightmost derivation in a, run back words.